Syllabus Details

Syllabus ID:	13201
Syllabus Name:	**Python for Applied Data Science _Lập trình Python cho khoa học dữ liệu ứng dụng**
Course Name English:	**Python for Applied Data Science**
Subject Code:	**PDS301m**
Learning-Teaching Method:	Inquiry-Based Learning,In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45 contact hours + 1h FE + 104h self-study
Pre-Requisite:	MAS291
Description:	This hands-on course introduces students to Python programming with a specific focus on data science applications. Designed for beginners, it starts with the essentials of programming and guides learners through structured modules including Python basics, data structures, programming fundamentals, data handling, and APIs. Students will learn to write clean and efficient Python code, manipulate and analyze datasets, and automate data collection from external sources. Each topic is reinforced with review sessions and lab-based activities to ensure practical understanding and application. By the end of the course, students will be able to develop Python programs that solve real-world data problems and form a strong foundation for further study in data science and machine learning.
StudentTasks:	- Student must complete the coursera course:https://www.coursera.org/learn/python-for-applied-data-science-ai/ - Students must attend more than 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU LMS at https://flm.fpt.edu.vn/ for up-to-date course information - Learn by coding practice
Tools:	Anacoda, Jupyter Notebook, Google Colab, Visual Studio Code
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

2 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Python for Data Science, AI & Development	Joseph Santarcangelo	Coursera	N/A	N/A	N/A	True	False	True	https://www.coursera.org/learn/python-for-applied-data-science-ai
2	Python Data Science Handbook	Jake VanderPlas	O'Reilly Media	2022	2nd	978-1-098121198	False	True	False	

5 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Develop a foundational understanding of Python programming by learning basic syntax, data types, expressions, variables, and string operations.
2	CLO2	Understand the key data structure in Python such as using data structures, conditions and branching, loops, functions, exception handling, objects, and classes.
3	CLO3	Know how to apply proficiency in using Python libraries such as Pandas and Numpy and developing code using Jupyter Notebooks.
4	CLO4	Understand and implement the way to access and extract web-based data by working with REST APIs using requests and performing web scraping
5	CLO5	Apply AI tools to enhance Python programming skills and critically justify the logic of the implemented code

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Introduction programming for Data Science + Set up environment	Offline	CLO1	ITU	Slide: Guide	PDS301m	Set up environment	
2	Introduction programming for Data Science + Set up environment	Offline	CLO1	ITU	Slide: Guide	PDS301m	Set up environment	
3	Assignment introduction	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	T	Assignment		Assignment	
4	Python Basic	Offline	CLO1	T	Slide and Module 1 (online course): Python Basics	link4-1	Complete module 1 and review slide: Python Basic	
5	Python Basic (con't)	Offline	CLO1	T	Slide and Module 1 (online course): Python Basics	link5-1	Complete module 1 and review slide: Python Basic	
6	Project introduction	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	T	Project		Project	
7	Lab 1 & Lab 2 evaluation	Offline	CLO1, CLO5	U	Lab 1 & Lab 2 tutorial		Do the lab and submit to system	
8	Lab 1 & Lab 2 evaluation	Offline	CLO1, CLO5	U	Lab 1 & Lab 2 tutorial		Do the lab and submit to system	
9	Python Data Structures: List & Tuples	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link9-1	Complete module 2 and review slide: Python Data Structures	
10	Python Data Structures: List & Tuples (con't)	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link10-1	Complete module 2 and review slide: Python Data Structures	
11	Python Data Structures: List & Tuples (con't)	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link11-1	Complete module 2 and review slide: Python Data Structures	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	Python Data Structures: Dictionaries	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link12-1	Complete module 2 and review slide: Python Data Structures	
13	Python Data Structures: Dictionaries (con't)	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link13-1	Complete module 2 and review slide: Python Data Structures	
14	Python Data Structures: Dictionaries (con't)	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link14-1	Complete module 2 and review slide: Python Data Structures	
15	Python Data Structures: Sets	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link15-1	Complete module 2 and review slide: Python Data Structures	
16	Python Data Structures: Sets (con't)	Offline	CLO2	T	Slide and Module 2 (online course): Python Data Structures	link16-1	Complete module 2 and review slide: Python Data Structures	
17	Review project -phase 1	Offline	CLO2	TU	Project		Present project	
18	Evaluation Lab 3-4-5 Evaluation	Offline	CLO2, CLO5	U	Lab 3-4-5 tutorial		Do the lab and submit to system	
19	Evaluation Lab 3-4-5 Evaluation	Offline	CLO2, CLO5	U	Lab 3-4-5 tutorial		Do the lab and submit to system	
20	Evaluation Lab 3-4-5 Evaluation	Offline	CLO2, CLO5	U	Lab 3-4-5 tutorial		Do the lab and submit to system	
21	Conditional & Branching	Offline	CLO2	T	Slide and Module 3 (online course): Conditional & Branching	link21-1	Complete module 3 (part 1) and review slide: Conditional & Branching	
22	Conditional & Branching (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Conditional & Branching	link22-1	Complete module 3 (part 1) and review slide: Conditional & Branching	
23	Conditional & Branching (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Conditional & Branching	link23-1	Complete module 3 (part 1) and review slide: Conditional & Branching	
24	Lab 6 Evaluation	Offline	CLO2, CLO5	U	Lab 6 tutorial		Do the lab and submit to system	
25	Loops	Offline	CLO2	T	Slide and Module 3 (online course): Loops	link25-1	Complete module 3 (part 1) and review slide: Loops	
26	Loops (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Loops	link26-1	Complete module 3 (part 1) and review slide: Loops	
27	Review project -phase 2	Offline	CLO2	TU	Project		Present project	
28	Lab 7 Evaluation	Offline	CLO2	U	Lab 7 tutorial		Do the lab and submit to system	
29	Functions	Offline	CLO2	T	Slide and Module 3 (online course): Functions	link29-1	Complete module 3 (part 1) and review slide: Functions	
30	Functions (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Functions	link30-1	Complete module 3 (part 1) and review slide: Functions	
31	Functions (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Functions	link31-1	Complete module 3 (part 1) and review slide: Functions	
32	Lab 8 Evaluation	Offline	CLO2, CLO5	U	Lab 8 tutorial		Do the lab and submit to system	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
33	Objects and Classes	Offline	CLO2	T	Slide and Module 3 (online course): Objects and Classes	link33-1	Complete module 3 (part 2) and review slide: Objects and Classes	
34	Objects and Classes (con't)	Offline	CLO2	T	Slide and Module 3 (online course): Objects and Classes	link34-1	Complete module 3 (part 2) and review slide: Objects and Classes	
35	Review project -phase 3	Offline	CLO2	TU	Project		Present project	
36	Lab 9 Evaluation	Offline	CLO2, CLO5	U	Lab 9 tutorial		Do the lab and submit to system	
37	Progress test 1	Offline	CLO2, CLO5	U	Progress test 1		Progress test 1	
38	Working with data in Python	Offline	CLO2	T	Slide and Module 4 (online course): Working with data in Python	link38-1	Complete module 4 and review slide: Working with data in Python	
39	Working with data in Python (con't)	Offline	CLO2	T	Slide and Module 4 (online course): Working with data in Python	link39-1	Complete module 4 and review slide: Working with data in Python	
40	Working with data in Python (con't)	Offline	CLO2	T	Slide and Module 4 (online course): Working with data in Python	link40-1	Complete module 4 and review slide: Working with data in Python	
41	Lab 10 Evaluation	Offline	CLO1, CLO2, CLO5	U	Lab 10 tutorial		Do the lab and submit to system	
42	APIs and Data collection	Offline	CLO3	T	Slide and Module 5 (online course): APIs and Data collection	link42-1	Complete module 5 and review silde: APIs and Data collection	
43	APIs and Data collection (con't)	Offline	CLO3	T	Slide and Module 5 (online course): APIs and Data collection	link43-1	Complete module 5 and review silde: APIs and Data collection	
44	APIs and Data collection (con't)	Offline	CLO3	T	Slide and Module 5 (online course): APIs and Data collection	link44-1	Complete module 5 and review silde: APIs and Data collection	
45	Lab 11 Evaluation	Offline	CLO3, CLO5	U	Lab 11 tutorial		Do the lab and submit to system	
46	Progress test 2	Offline	CLO3, CLO5	U	Progress test 2		Progress test 2	
47	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	
48	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	
49	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	
50	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
51	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	
52	Assignment evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Assignment tutorial		Do assignment and submit to system	
53	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
54	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
55	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
56	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
57	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
58	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
59	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	
60	Group project presentation & evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U	Proposed project based on project guidelines		Do project, present and submit to system	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	20.0%	1	At home	CLO1, CLO2, CLO3, CLO4, CLO5	Homework, mini-projects, and optionally the constructivist approach as proposed by the lecturer.	by instructor	All topics	Individual or team work, guided by instructor, submission by a given deadline	Instructor has responsible for guiding students in carrying out the task View assignment on slot 20

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Labs	on-going	11	10.0%	1		CLO1, CLO2, CLO3, CLO4, CLO5	Mini-project	by instructor	All topics	Individual or team work, guided by instructor, submission by a given deadline	Instructor has resposibility to guide and review the lab practice for students.
3	Progress test 1	on-going	1	5.0%	>0	30'	CLO1, CLO2,	Multiple choices	25	CLO1, CLO2, CLO5	in class, by instructor	N/A
4	Progress test 2	on-going	1	5.0%	>0	30'	CLO3, CLO4,	Multiple choices	25	CLO3, CLO4, CLO5	in class, by instructor	N/A
5	Project	on-going	1	30.0%	4	360'	CLO1, CLO2, CLO3, CLO4, CLO5	Project	by Instructor	All topics; knowledge and skills	The examination is conducted by the exam board. A minimum of two teachers is required for grading the Final Project. Each team is allocated 30 minutes for the presentation of their project followed by a Q&A session. Evaluation of the final presentation is based on the final report, project demonstration, and Q&A interaction. The grade is determined by averaging the grades given by the two teachers.	
6	Final exam	Final exam	1	30.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5	Multiple choices Marked by Computer	50	All topics	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.